Lecture 5

CSS3223 Calculus

CHAPTER 3 : DIFFERENTIATION

3.1 THE DERIVATIVE AS A FUNCTION
3.2 DIFFERENTIATION RULES
3.3 THE DERIVATIVE AS A RATE OF CHANGE
3.4 DERIVATIVES OF TRIGONOMETRIC FUNCTIONS
3.5 THE CHAIN RULE AND PARAMETRIC EQUATIONS
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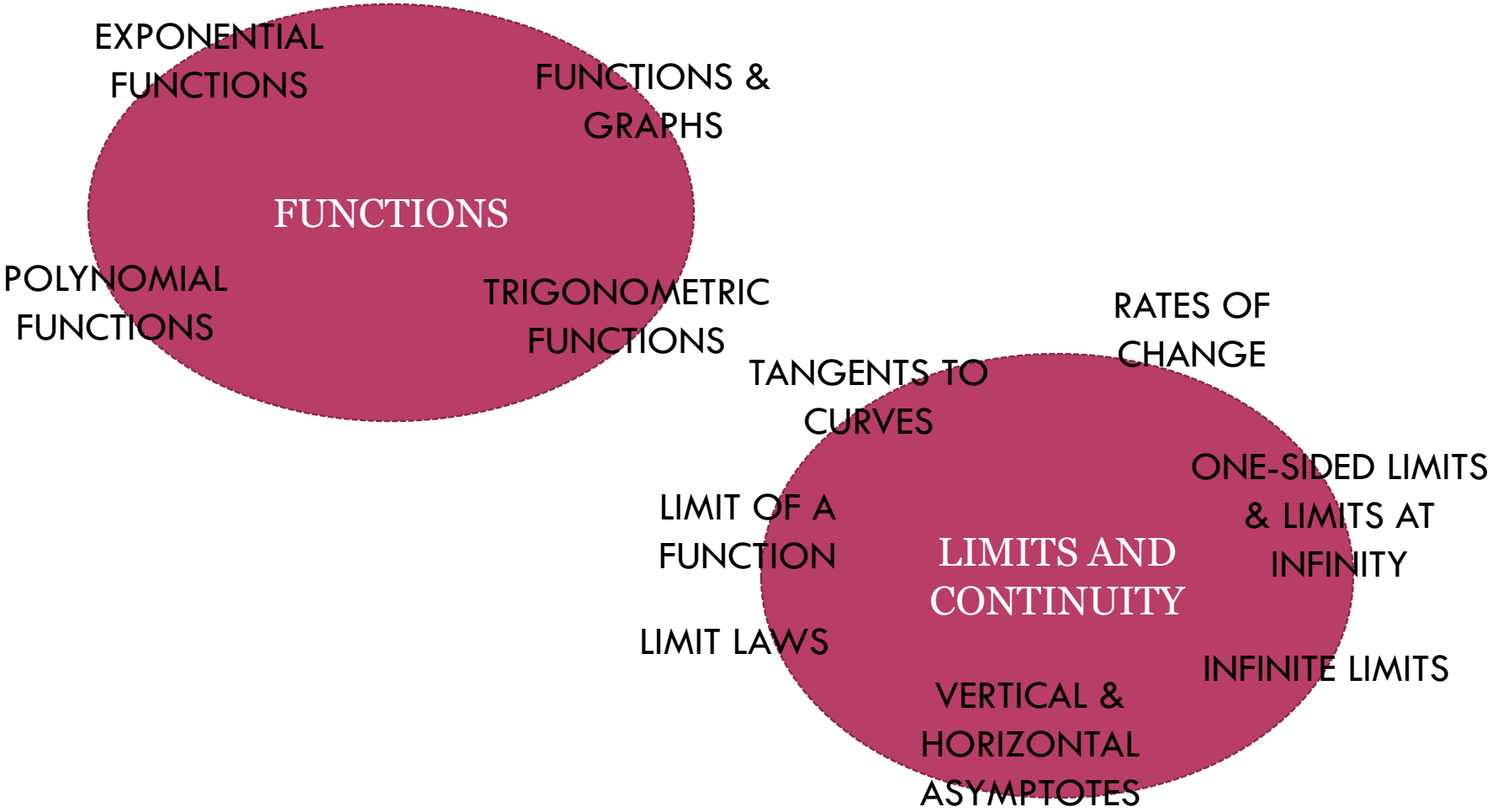
3.7 DERIVATIVES OF INVERSE FUNCTIONS AND LOGARITHMS
3.8 INVERSE TRIGONOMETRIC FUNCTIONS
3.9 RELATED RATES
3.10 LINEARIZATION AND DIFFERENTIALS
3.11 HYPERBOLIC FUNCTIONS

Recollection

- We have seen:
  - Functions
    - What they are
    - How are they pictured as graphs
    - How they are combined
    - How they are transformed
    - How they can be classified
  - Limits and Continuity
    - It describes the way a function varies
      - Some function varies continuously
      - Others can have values that jump, vary erratically, or increase or decrease without bound

Recollection



Derivative as a Function

- Objectives:
  - To recall what derivative is
  - To understand that derivative of a function is a function itself
  - To understand the terms: differentiable and differentiation
  - To be able to calculate a derivative using first principle

Derivative of the function $f(x)$

DEFINITION The **derivative** of the function $f(x)$ with respect to the variable x is the function f' whose value at x is

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h},$$

provided the limit exists.

- If f' exists at particular point x, we say that f is differentiable (has a derivative) at x.
- If f' exists at every point in the domain of f, we call f differentiable.

Alternative formula for the derivative

- If we write $z = x + h$, then $h = z - x$ and h approaches 0 if and only if z approaches x. Therefore, an equivalent definition of the derivative is as follows

Alternative Formula for the Derivative

$$f'(x) = \lim_{z \to x} \frac{f(z) - f(x)}{z - x}.$$

Calculating derivative from the definition
(first principle)

- The process of calculating a derivative is called differentiation.
- We use notation $\frac{d}{dx}f(x)$ as another way to denote the derivative $f'(x)$.

Example 1

1. Find the derivative of $y = \sqrt{x}$ for $x > 0$.

$$y = \sqrt{x} = x^{\frac{1}{2}}.$$

$$y' = \frac{1}{2}x^{-\frac{1}{2}}.$$

2. Using alternative formula, find the derivative of $f(x) = \dfrac{x}{x-1}$

$$f'(x) = \lim_{(x+h)\to x} \frac{f(x+h)-f(x)}{x+h-x}$$

$$= \lim_{(x+h)\to x} \frac{\dfrac{x+h}{x+h-1} - \dfrac{x}{x-1}}{x+h-x}$$

$$= \lim_{(x+h)\to x} \frac{1}{h}\left(\frac{(x+h)(x-1)-x(x+h-1)}{(x+h-1)(x-1)}\right)$$

$$= \lim_{(x+h)\to x} \frac{1}{h}\left(\frac{-h}{(x+h-1)(x-1)}\right)$$

$$= \lim_{(x+h)\to x}\left(\frac{-1}{(x+h-1)(x-1)}\right)$$

$$= \frac{-1}{(x-1)^2}$$

Notations

- Many ways to denote the derivative of a function $y = f(x)$, where the independent variable is x and dependent variable is y.
- Common ones:

$$f'(x) = y' = \frac{dy}{dx} = \frac{df}{dx} = \frac{d}{dx} f(x) = D(f)(x) = D_x f(x).$$

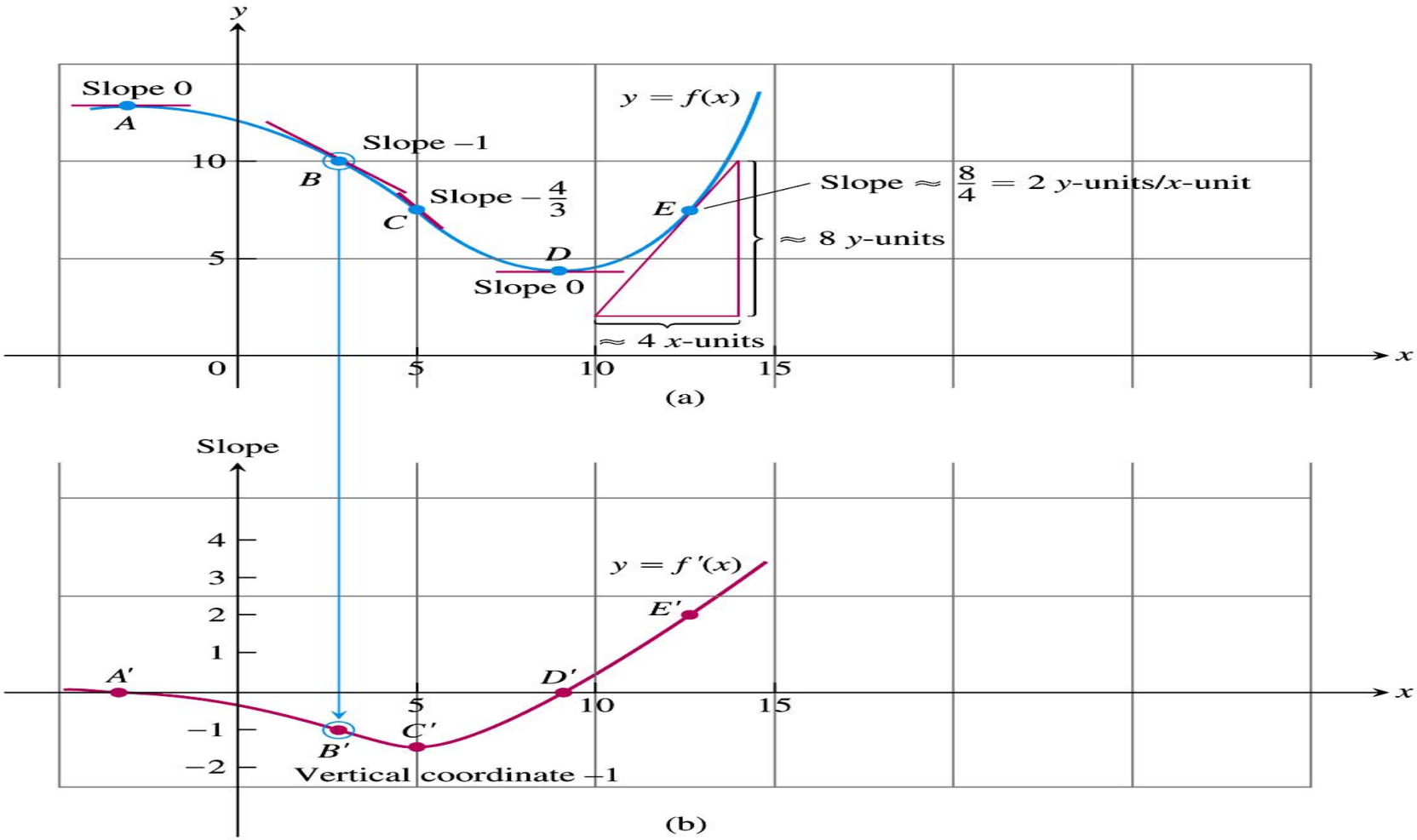
- To indicate the value of a derivative at a specified number $x = a$, we use

$$f'(x) = \left.\frac{dy}{dx}\right|_{x=a} = \left.\frac{df}{dx}\right|_{x=a} = \left.\frac{d}{dx} f(x)\right|_{x=a}$$

Graphing the derivative

- We can make a reasonable plot of the derivative $y=f(x)$ by estimating the slopes on the graph of f.
- We plot the points $(x,f'(x))$ in the xy-plane and connect them with a smooth curve, which represents $y = f'(x)$.

The graph of $y = f'(x)$



Differentiable on an interval

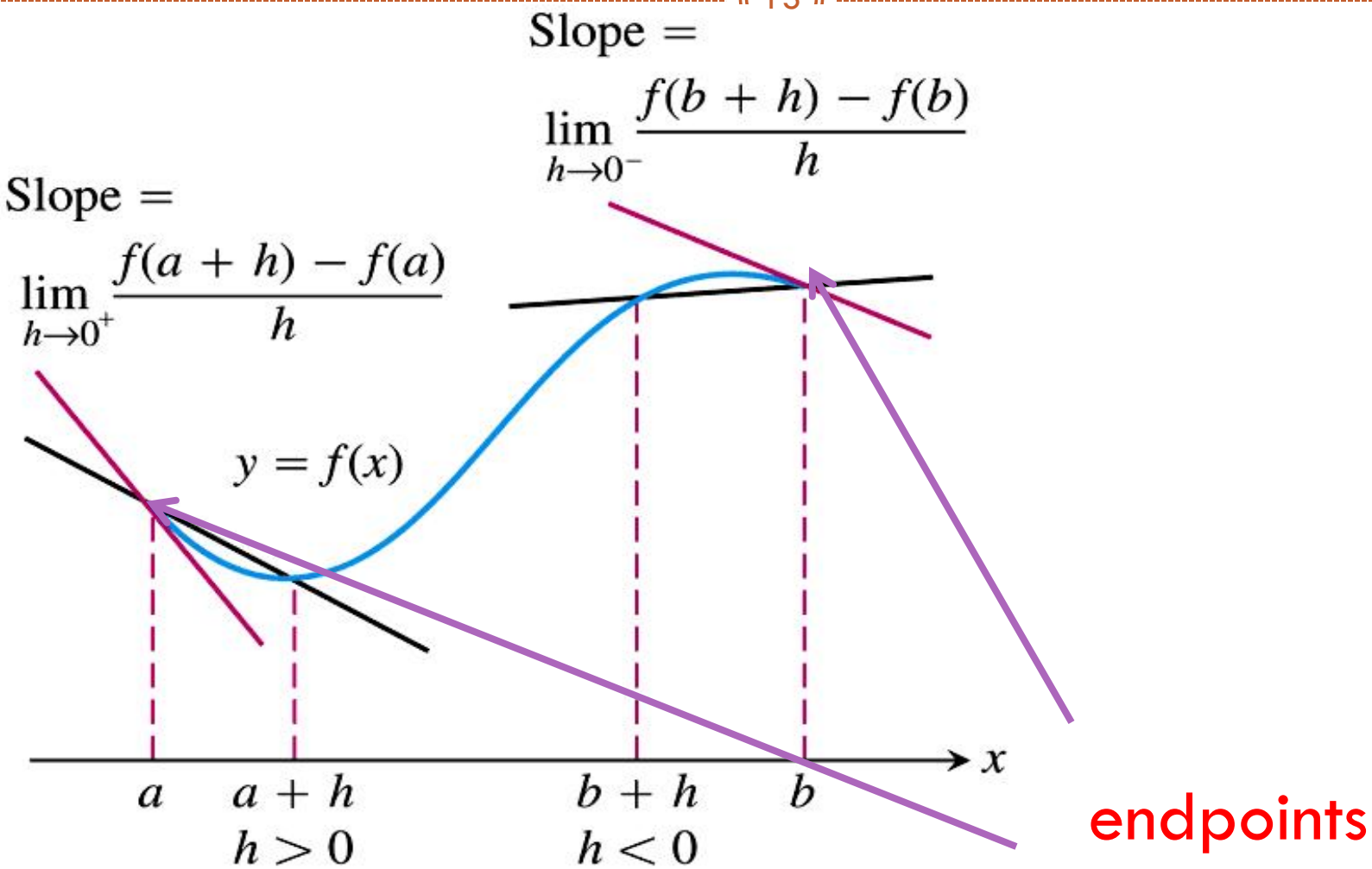
- A function $y = f(x)$ is differentiable on an open interval (finite or infinite) if it has a derivative at each point of the interval.
- It is differentiable on a closed interval $[a,b]$ if it is differentiable on the interior (a,b) and if the limits

$$\lim_{h \to 0^+} \frac{f(a+h) - f(a)}{h}$$ Right-hand derivative at a

$$\lim_{h \to 0^-} \frac{f(b+h) - f(b)}{h}$$ Left-hand derivative at b

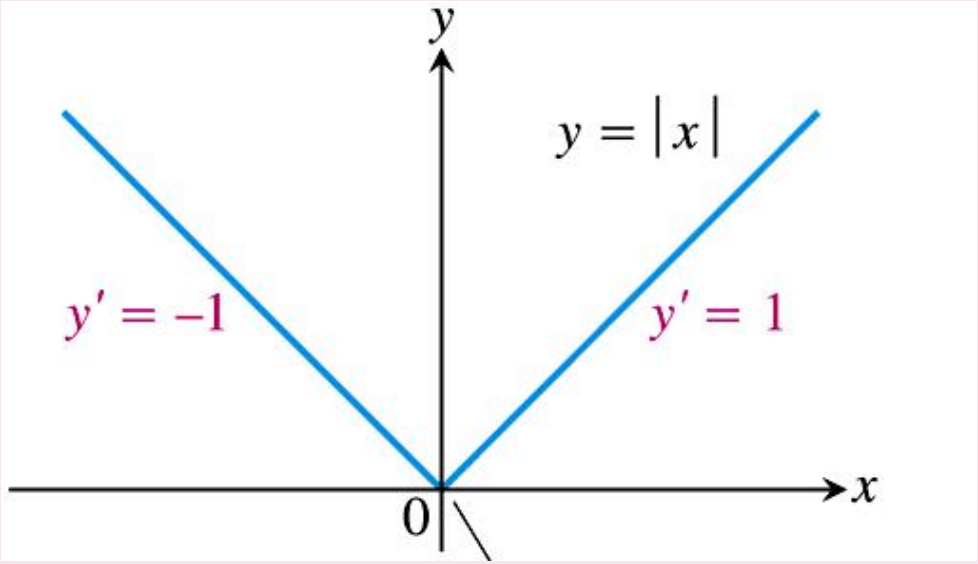
exist at the endpoints .

Derivative at endpoints are one-sided limits



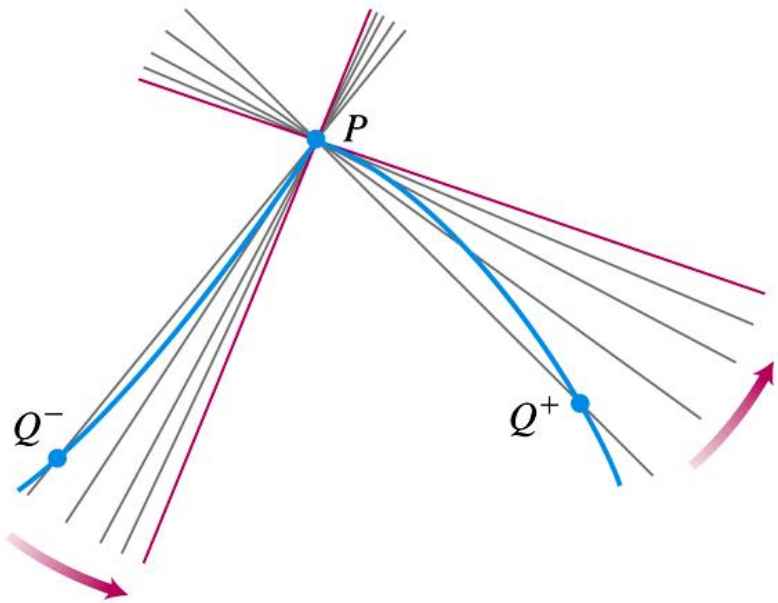
Example 2

- Show that the function $y = |x|$ is differentiable on $(-\infty,0)$ and $(0,\infty)$ but has no derivative at $x = 0$

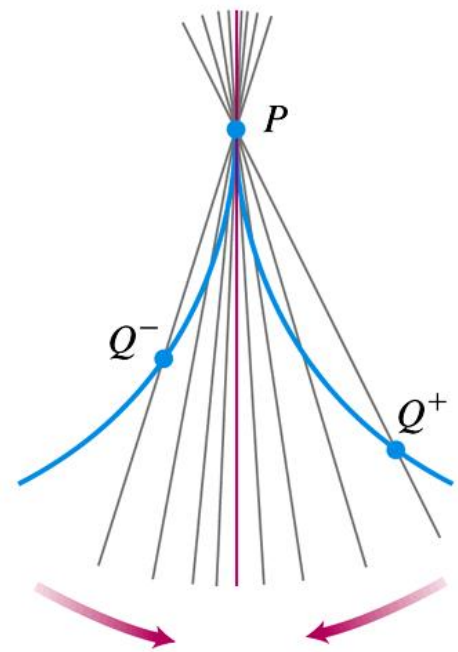


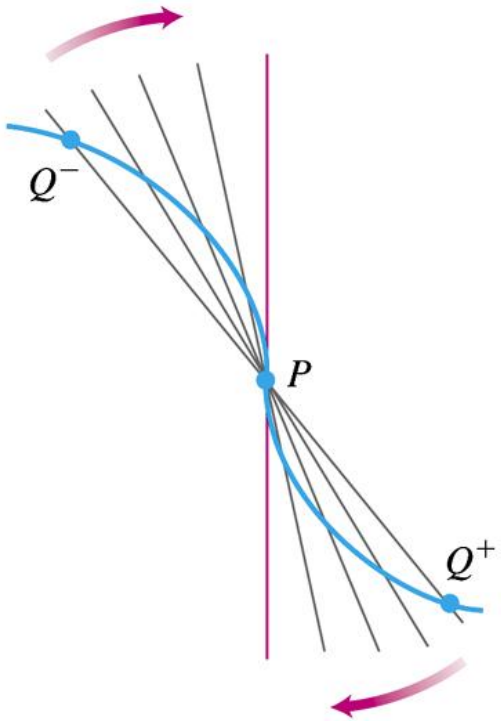
When does a function NOT HAVE a derivative at a point?

1. a *corner*, where the one-sided derivatives differ.

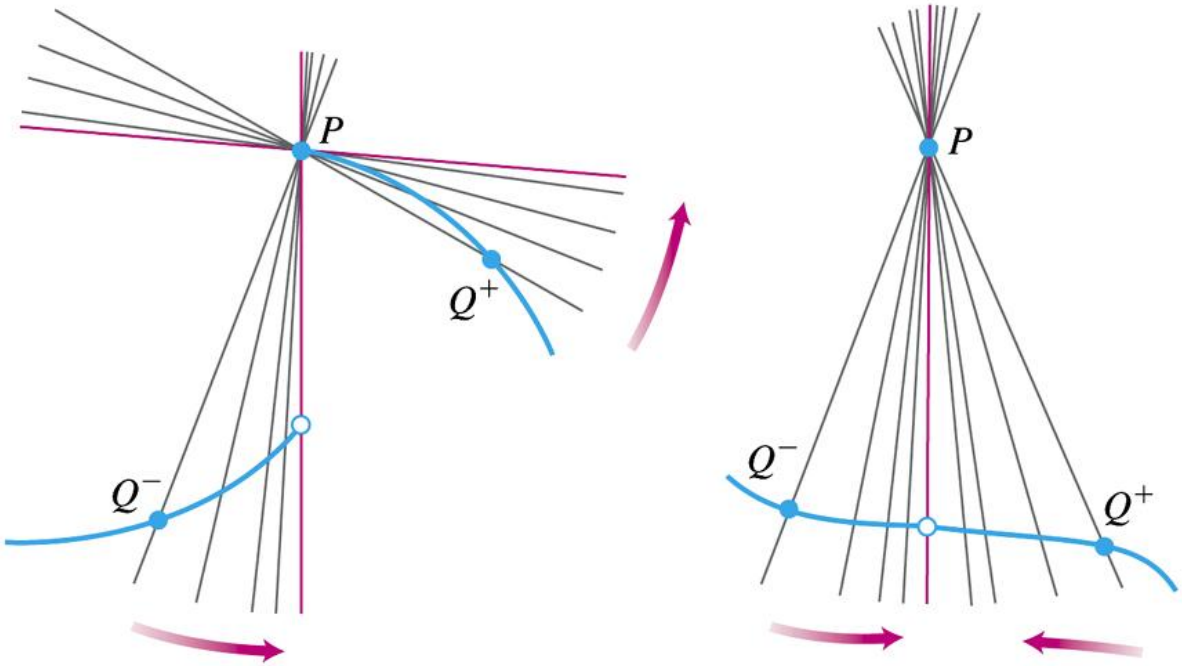


2. a *cusp*, where the slope of PQ approaches ∞ from one side and $-\infty$ from the other.





3. a *vertical tangent*, where the slope of PQ approaches ∞ from both sides or approaches $-\infty$ from both sides (here, $-\infty$).



4. a *discontinuity* (two examples shown).

Differentiable functions are continuous

- A function is continuous at every point where it has a derivative.

THEOREM 1 Differentiability Implies Continuity

If f has a derivative at $x = c$, then f is continuous at $x = c$.

Summary

- We learned about:
  - The derivative of a function
  - How to calculate derivatives from first principle
  - Derivative of a function is a function itself and thus can be graphed
  - What it means by differentiable
  - When a function does not have a derivative at a function
  - Differentiable functions are actually continuous

Recollection

- We have seen:
  - Derivative as a function
    - What it means by derivative, differentiable, differentiation
    - How to find the derivative from definition
    - Knowing that derivative of a function is a function itself
    - How to sketch the graph of the derivative of a function
    - Knowing that differentiable functions are continuous

Differentiation rules for polynomials, exponentials, products and quotients

- Objectives:
  - To introduce several rules that allow us to differentiate
    - Constant functions
    - Power functions
    - Polynomials
    - Exponential functions
    - Rational functions
    - Combinations of some of the functions above

Derivative of a constant function

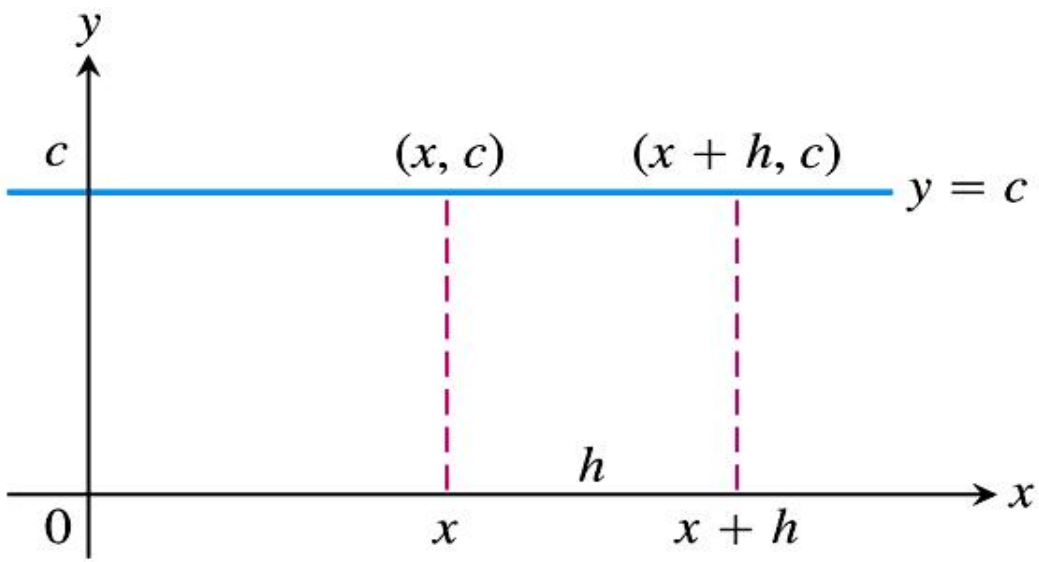
- The first rule of differentiation is that the derivative of every constant function is zero

RULE 1 Derivative of a Constant Function

If f has the constant value $f(x) = c$, then

$$\frac{df}{dx} = \frac{d}{dx}(c) = 0.$$

What does $(d/dx)c = 0$ means?



Notice:

- The curve of a constant function is horizontal
- The slope of a horizontal line is zero everywhere
- This means that the values of constant functions never change

Power rule

RULE 2 Power Rule (General Version)

If n is any real number, then

$$\frac{d}{dx}x^n = nx^{n-1},$$

for all x where the powers x^n and x^{n-1} are defined.

- It is valid for every real numbers of n.
- To apply, we subtract 1 from the original exponent n, then multiply the result by n.

Example 1

Differentiate the following functions:

i) x^4 $\qquad \dfrac{d}{dx}\left(x^4\right) = 4x^3$

ii) $\sqrt[5]{x}$ $\qquad \sqrt[5]{x} = x^{\frac{1}{5}}.$

$$\frac{d}{dx}\left(\sqrt[5]{x}\right) = \frac{1}{5}x^{-\frac{4}{5}}$$

Constant Multiple Rule

RULE 3 Constant Multiple Rule

If u is a differentiable function of x, and c is a constant, then

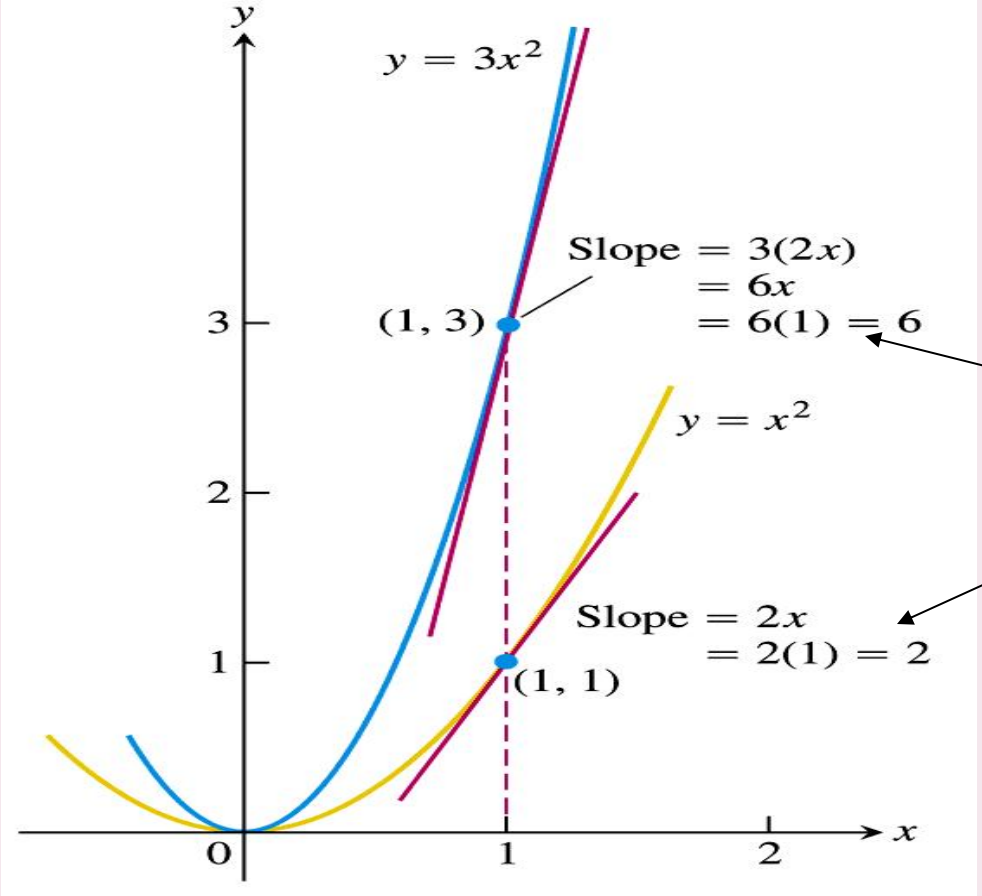
$$\frac{d}{dx}(cu) = c\frac{du}{dx}.$$

- If n is real number then,

$$\frac{d}{dx}(cx^n) = ? \qquad \frac{d}{dx}\left(cx^n\right) = cnx^{n-1}.$$

Example 2

- Differentiate $f(x) = 3x^2$



Value of slope for the curve at x = 1

Derivative sum rule

RULE 4 Derivative Sum Rule

If u and v are differentiable functions of x, then their sum $u + v$ is differentiable at every point where u and v are both differentiable. At such points,

$$\frac{d}{dx}(u + v) = \frac{du}{dx} + \frac{dv}{dx}.$$

<u>Example</u>

Differentiate $y = x^{4/5} + 5x$

$$\frac{d}{dx}(y) = \frac{d}{dx}\left(x^{4/5} + 5x\right) = \frac{d}{dx}\left(x^{\frac{4}{5}}\right) + \frac{d}{dx}(5x) = \frac{4}{5}x^{\frac{-1}{5}} + 5.$$

Derivative of the natural exponential function

Derivative of the Natural Exponential Function

$$\frac{d}{dx}(e^x) = e^x$$

<u>Example</u>

Differentiate the function $f(x) = ae^x$

$$\frac{d}{dx}\left(ae^x\right) = ae^x.$$

Products and Quotients

RULE 5 Derivative Product Rule

If u and v are differentiable at x, then so is their product uv, and

$$\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}.$$

RULE 6 Derivative Quotient Rule

If u and v are differentiable at x and if $v(x) \neq 0$, then the quotient u/v is differentiable at x, and

$$\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v\dfrac{du}{dx} - u\dfrac{dv}{dx}}{v^2}.$$

Example

- Differentiate the following functions:

1. $y = (x^3 + \frac{1}{2})(x^4 + 3)$

$$\frac{d}{dx}\left((x^3+\frac{1}{2})(x^4+3)\right)$$

$$=\left[(x^3+\frac{1}{2})\frac{d}{dx}(x^4+3)\right]+\left[(x^4+3)\frac{d}{dx}(x^3+\frac{1}{2})\right]$$

$$=(x^3+\frac{1}{2})4x^3+(x^4+3)3x^2$$

$$=7x^6+2x^3+9x^2.$$

2. $y = \frac{t^3 - 1}{t^2 + 5}$

$$u=\left(t^3-1\right), v=\left(t^2+5\right), \frac{du}{dx}=3t^2, \frac{dv}{dx}=2t.$$

$$\frac{d}{dx}y=\frac{\left(t^2+5\right)(3t^2)-\left(t^3-1\right)(2t)}{\left(t^2+5\right)^2}=\frac{3t^4+15t^2-2t^4+2t}{\left(t^2+5\right)^2}=\frac{t^4+15t^2+2t}{\left(t^2+5\right)^2}.$$

Second- and Higher-Order Derivatives

- If $y = f(x)$ is a differentiable function, then its derivative $f'(x)$ is also a function.
- If $f'(x)$ is also differentiable, then we can differentiate $f'(x)$ to get a new function of x denoted by $f''(x)$.
- Hence, $f''(x) = (f')'$.
- This function is called the second derivative of f.

$$f''(x) = y'' = \frac{d^2 y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{dy'}{dx} = D^2(f)(x) = D_x{}^2 f(x).$$

Example

- Find the first and second derivatives of y:

$$y = 3x^4 - 0.74x^{2/3} + 180$$

ANSWERS:

$$1^{\text{st}}\ \text{derivative}: 12x^3 - \frac{2}{3}(0.74)x^{-\frac{1}{3}}.$$

$$2^{\text{rd}}\ \text{derivative}: 36\text{x}^2 + \frac{2}{9}(0.74)x^{-\frac{4}{3}}.$$

Derivatives of Trigonometric Functions

- Motivation:
  - Many real problems are periodic in nature:
    - Electromagnetic fields, heart rhythms, tides, weather, etc.
  - The derivatives of sines and cosines play a key role in describing periodic changes

- Objectives:
  - To be able to differentiate the six basic trigonometric functions

Derivative of the Sine and Cosine Functions

The derivative of the sine function is the cosine function:

$$\frac{d}{dx}(\sin x) = \cos x.$$

The derivative of the cosine function is the negative of the sine function:

$$\frac{d}{dx}(\cos x) = -\sin x$$

Example

- Differentiate these functions

i) $y = x^2 \sin x$

$$\frac{d}{dx}(x^2)\sin x + x^2 \frac{d}{dx}(\sin x) = 2x\sin x + x^2 \cos x$$

ii) $y = \dfrac{\cos x}{x^3}$

$$-\frac{\sin x}{x^3} - \frac{3\cos x}{x^4} = -\frac{x\sin x + 3\cos x}{x^4}$$

iii) $y = 2(\sin x)(\cos x)^3$

$$2\frac{d}{dx}[\sin(x)\cos^3(x)]$$
$$= 2\left[\sin(x)\left(3\cos^2(x)\frac{d}{dx}\cos(x)\right) + \cos^3(x)\cos(x)\right]$$
$$= 2[\sin(x)(3\cos^2(x)[-\sin x]) + \cos^3(x)\cos(x)]$$
$$= 2\cos^4(x) - 6\cos^2(x)\sin^2(x)$$

Derivatives of the other basic trigonometric functions

- Recall that the other basic trigonometric functions has relations with sine and cosine, i.e.:

$$\tan x = \frac{\sin x}{\cos x} \quad \sec x = \frac{1}{\cos x} \quad \cot x = \frac{\cos x}{\sin x} \quad \csc x = \frac{1}{\sin x}$$

- Thus, their derivatives can be obtained from the Quotient rule:

Derivatives of the Other Trigonometric Functions

$$\frac{d}{dx}(\tan x) = \sec^2 x \qquad \frac{d}{dx}(\cot x) = -\csc^2 x$$

$$\frac{d}{dx}(\sec x) = \sec x \tan x \qquad \frac{d}{dx}(\csc x) = -\csc x \cot x$$

Example

- Find y'' if $y = \sec x$

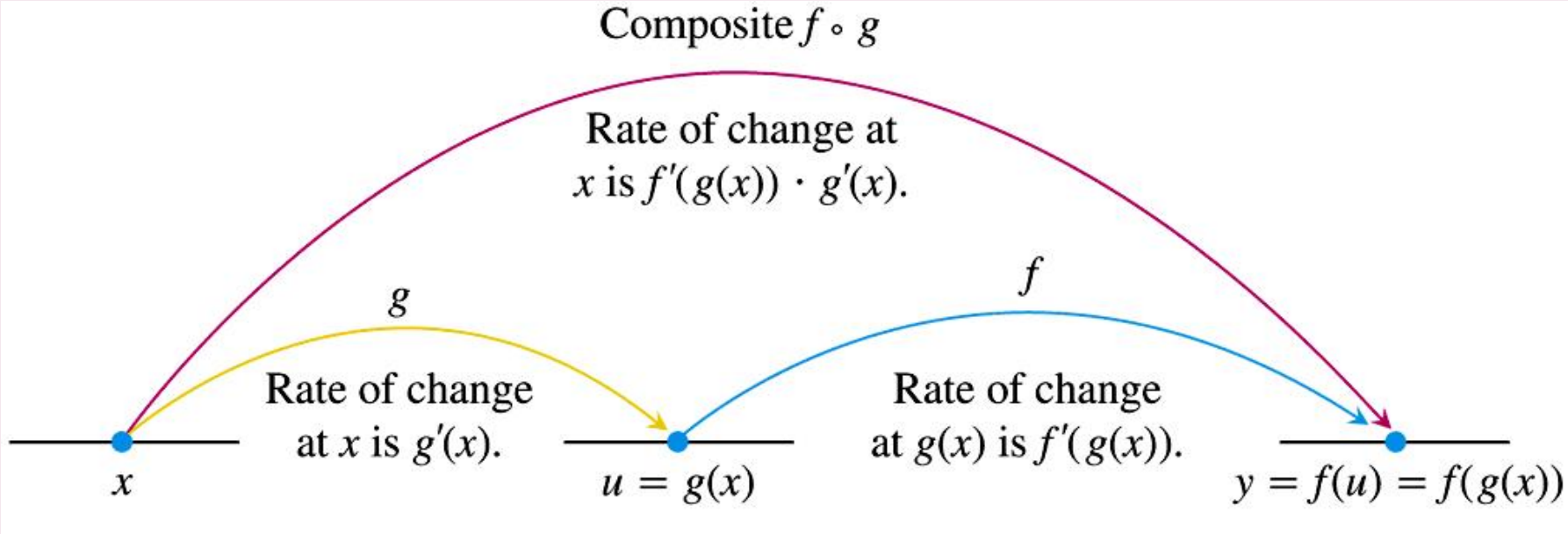
$$y' = \sec x \tan x$$

$$y'' = \frac{d}{dx}(\sec x \tan x) = \sec x \tan^2 x + \sec^3 x$$

The Chain Rule

- Objectives:
  - To recall what composite functions are
  - To be able to differentiate a composite functions
    - The purpose of chain rule

Composite functions



Rates of change multiply: The derivative of $f \circ g$ at x is the derivative of f at $g(x)$ times the derivative of g at x

The Chain Rule

THEOREM 3 The Chain Rule

If $f(u)$ is differentiable at the point $u = g(x)$ and $g(x)$ is differentiable at x, then the composite function $(f \circ g)(x) = f(g(x))$ is differentiable at x, and

$$(f \circ g)'(x) = f'(g(x)) \cdot g'(x).$$

In Leibniz's notation, if $y = f(u)$ and $u = g(x)$, then

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx},$$

where dy/du is evaluated at $u = g(x)$.

Simple Example

<u>Example of a composite function</u>

$$y = \sqrt{3x}$$

The function is the composition of the functions

$$y = \sqrt{u} \qquad \text{and} \qquad u = 3x$$

$$\frac{dy}{du} = \frac{1}{2\sqrt{u}} = \frac{1}{2\sqrt{3x}}, \quad \frac{du}{dx} = 3, \quad \therefore \frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx} = \frac{3}{2\sqrt{3x}}$$

"Outside-Inside" Rule

If $y = f(g(x))$, then

$$\frac{dy}{dx} = f'(g(x)) \cdot g'(x)$$

or, in words, differentiate the "outside" function f and evaluate it at the "inside" function $g(x)$ left alone; then multiply by the derivative of the "inside function".

Example

1. Differentiate $\sin(x^2 + e^x)$ with respect to x

$$= \cos\left(e^x + x^2\right) \cdot \frac{d}{dx}\left[e^x + x^2\right]$$

$$= \cos\left(e^x + x^2\right)\left(\frac{d}{dx}\left[e^x\right] + \frac{d}{dx}\left[x^2\right]\right)$$

$$= \cos\left(e^x + x^2\right)\left(e^x + 2x\right)$$

2. Differentiate $y = e^{x^3}$

$$= e^{x^3} \cdot \frac{d}{dx}\left[x^3\right]$$

$$= e^{x^3} \cdot 3x^2$$

The chain rule brings the formula for natural exponential functions

$$\frac{d}{dx}e^u = e^u\frac{du}{dx}.$$

<u>Example</u>

Differentiate $y = e^{\cos x}$

$$\frac{dy}{dx} = (e^{\cos x})(-\sin x) =- e^{\cos x}\sin x$$

Repeated use of the chain rule

- We sometimes use the Chain Rule two or more times to find a derivative.
- <u>Example</u>
- Find the derivative of $g(t) = \tan(5 - \sin 2t)$

$$= -\frac{\mathrm{d}}{\mathrm{d}t}[\tan(\sin(2t) - 5)]$$

$$= -\sec^2(\sin(2t) - 5) \cdot \frac{\mathrm{d}}{\mathrm{d}t}[\sin(2t) - 5]$$

$$-\sec^2(\sin(2t) - 5)\left(\frac{\mathrm{d}}{\mathrm{d}t}[\sin(2t)] + \frac{\mathrm{d}}{\mathrm{d}t}[-5]\right)$$

$$= -\sec^2(\sin(2t) - 5)\left(\cos(2t) \cdot \frac{\mathrm{d}}{\mathrm{d}t}[2t] + 0\right)$$

$$= -\sec^2(\sin(2t) - 5)\cos(2t) \cdot 2 \cdot \frac{\mathrm{d}}{\mathrm{d}t}[t]$$

$$= -2\sec^2(\sin(2t) - 5)\cos(2t) \cdot 1$$

$$= -2\cos(2t)\sec^2(\sin(2t) - 5)$$

Chain Rule with Powers of a Function

- If $f(u) = u^n$ and u is a differentiable function of x, then substituting $y = f(u)$ into the Chain Rule formula leads to the **Power Chain Rule:**

$$\frac{dy}{dx}u^n = nu^{n-1} \cdot \frac{du}{dx}$$

Example

- Find $\dfrac{d}{dx}\left(\dfrac{1}{3x-2}\right)$

$$= -\frac{\frac{\mathrm{d}}{\mathrm{d}x}[3x-2]}{(3x-2)^2}$$

$$= -\frac{3\cdot\frac{\mathrm{d}}{\mathrm{d}x}[x]+\frac{\mathrm{d}}{\mathrm{d}x}[-2]}{(3x-2)^2}$$

$$= -\frac{3\cdot 1+0}{(3x-2)^2}$$

$$= -\frac{3}{(3x-2)^2}$$

Summary

- We learned about:
  - Differentiation rules
  - How to find derivatives of higher order
  - How derivatives is used in motion along a line
  - Calculating the derivatives of the six basic trigonometric functions
  - Calculating the derivatives of composite functions
    - The chain rule

Recollection

- We have seen:
  - Derivative as a function
  - Differentiation rules
  - Derivative as a rate of change
  - Derivatives of trigonometric functions
  - The chain rule
- All the above is for the functions described in the form $y = f(x)$
  - y is expressed explicitly in terms of the variable x

Implicit Differentiation

- Objectives:
  - To understand what it means by implicit functions
  - To learn the technique involves in differentiating equation that defines implicit relation between the unknown function and the independent variable

Implicitly defined functions

- Explicitly defined functions:
  - Functions that can be written $y = f(x)$
  - Example: $y = x^2 + 3x, \quad y = \sqrt{x^2 + 2x^3}, \quad y = \dfrac{x}{\ln(x-1)}$
- **Implicitly** defined functions are the functions such that you **can only** write it in this way $\boldsymbol{F(x, y) = 0}$
  - Example:

$$x^3 + y^3 - 9xy = 0, \qquad \sin xy + x^3 + y^2 = 0, \qquad x^2 y + xy^3 = 9$$

Implicit Differentiation

Implicit Differentiation

1. Differentiate both sides of the equation with respect to x, treating y as a differentiable function of x.
2. Collect the terms with dy/dx on one side of the equation.
3. Solve for dy/dx.

Example

1) Differentiate

$$x^3 + y^3 - 9xy = 0$$

$Ans: 3x^2 + 3y^2\frac{dy}{dx} - 9\left(x\frac{dy}{dx} + y\right) = 0$

$3x^2 - 9y = 9x\frac{dy}{dx} - 3y^2\frac{dy}{dx}$

$\frac{3x^2 - 9y}{9x - 3y^2} = \frac{dy}{dx} \rightarrow \frac{dy}{dx} = \frac{x^2 - 3y}{3x - y^2}$

2) Find $\frac{d^2y}{dx^2}$ if $2x^3 - 3y^2 = 8$

$6x^2 - 6y\frac{d}{dx}(y) = 0$

$\frac{dy}{dx} = \frac{6x^2}{6y} = \frac{x^2}{y}$

$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{x^2}{y}\right) = \frac{y.2x - x^2\frac{dy}{dx}}{y^2} = \frac{2xy - x^2\frac{dy}{dx}}{y^2} =$

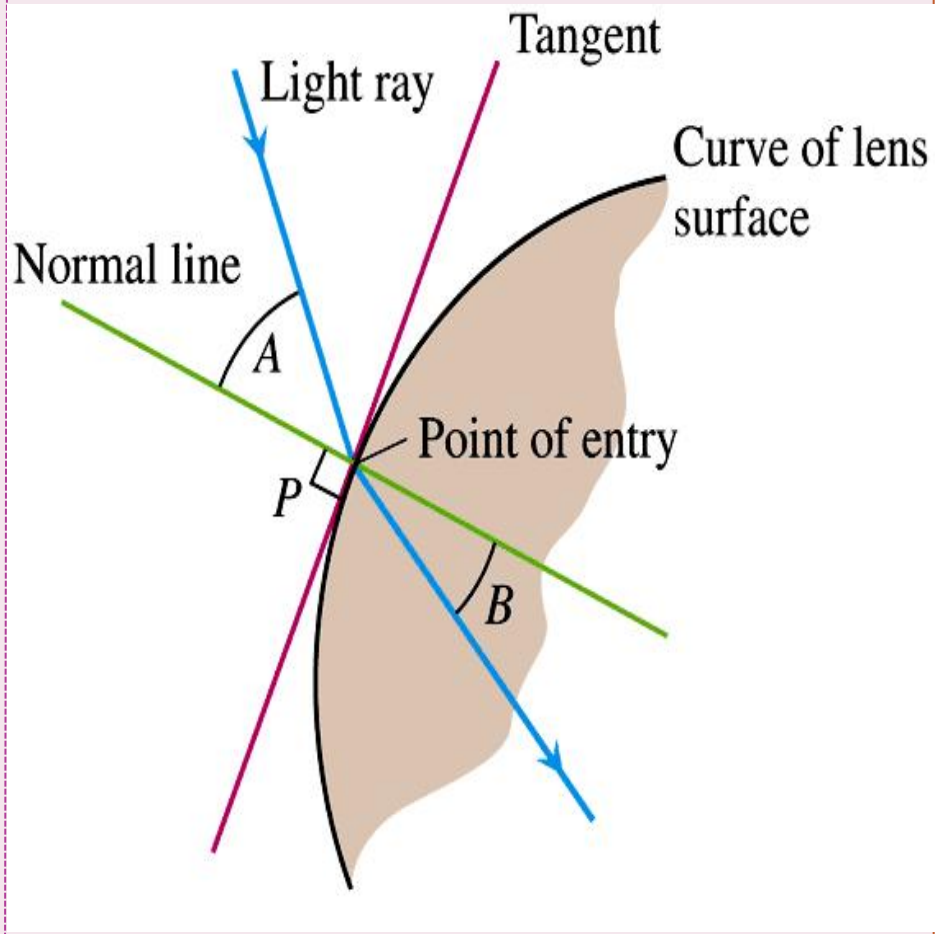
$\frac{2x}{y} - \frac{x^2}{y^2}\frac{dy}{dx} = \frac{2x}{y} - \frac{x^2}{y^2}\frac{x^2}{y} = \frac{x}{y}\left(2 - \frac{x^3}{y^2}\right)$

The Difference between Explicit and Implicit Equations

- An explicit equations has y as the subject and just x's on the other side. Eg. $y = x^2 - 4x + 2$
- An implicit equation has x's and y's mixed throughout the equation. Eg $y^3 + x^2 + 5y^2 - 2xy = 2$.

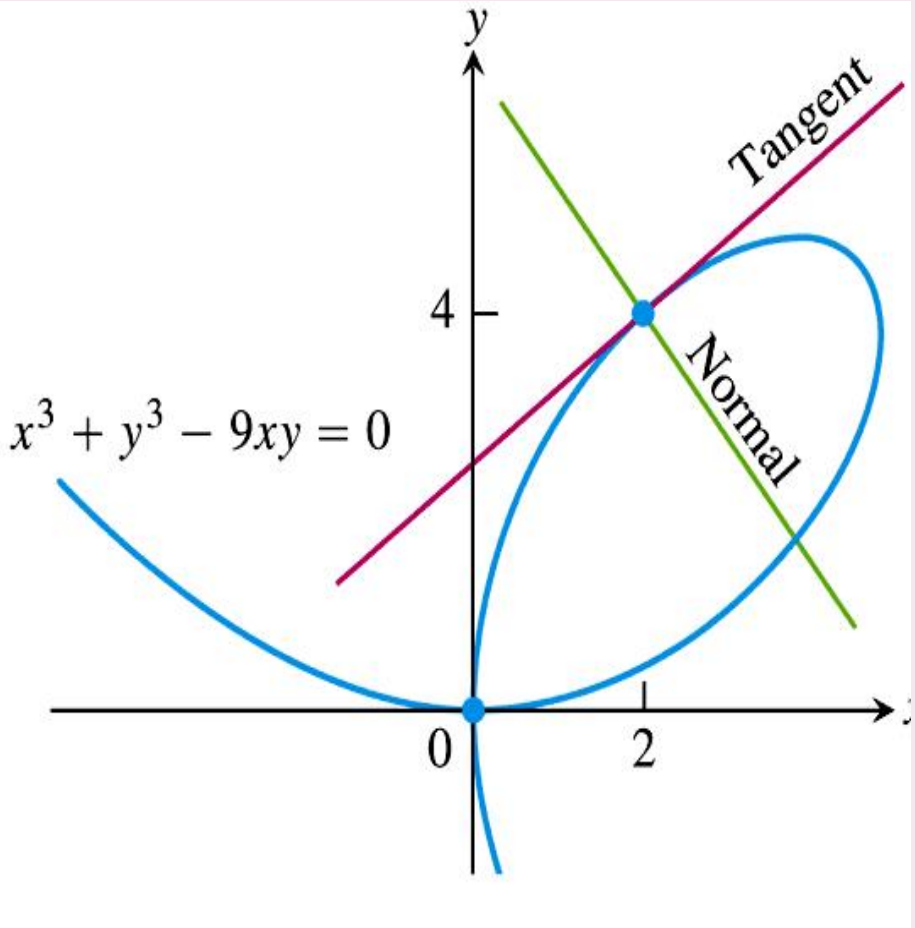
Lenses, Tangents, and Normal Lines

- Figure shows the profile of a lens, showing the bending (refraction) of a ray of light as it passes through the lens surface.



Example

Show that the point (2,4) lies on the curve

$$x^3 + y^3 - 9xy = 0$$

Then find the tangent and normal to the curve there.